

7(a)	correct grid shape (of wire)	B1
	fine wire / foil strip	B1
	plastic / insulating envelope containing the wire	B1
7(b)(i)	$2.00 / 6.00 = 153.0 / (R + 153.0)$ <i>or</i> $4.00 / 6.00 = R / (R + 153.0)$ (so $R = 306.0$)	C1
	$\Delta R = 306.0 - 300.0 = 6.0 \text{ } (\Omega)$	C1
	so $\Delta L = 8(.0) \times 10^{-5} \text{ m}$	A1

Question	Answer	Marks
7(b)(ii)	R or ΔR increases	B1
	$V^+ < V^-$ or $V_A < 2.00$ or V^+ / V_A decreases	M1
	output is negative / -5 V	A1
	diode X emits light / is 'on'	A1

Question	Answer	Marks
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